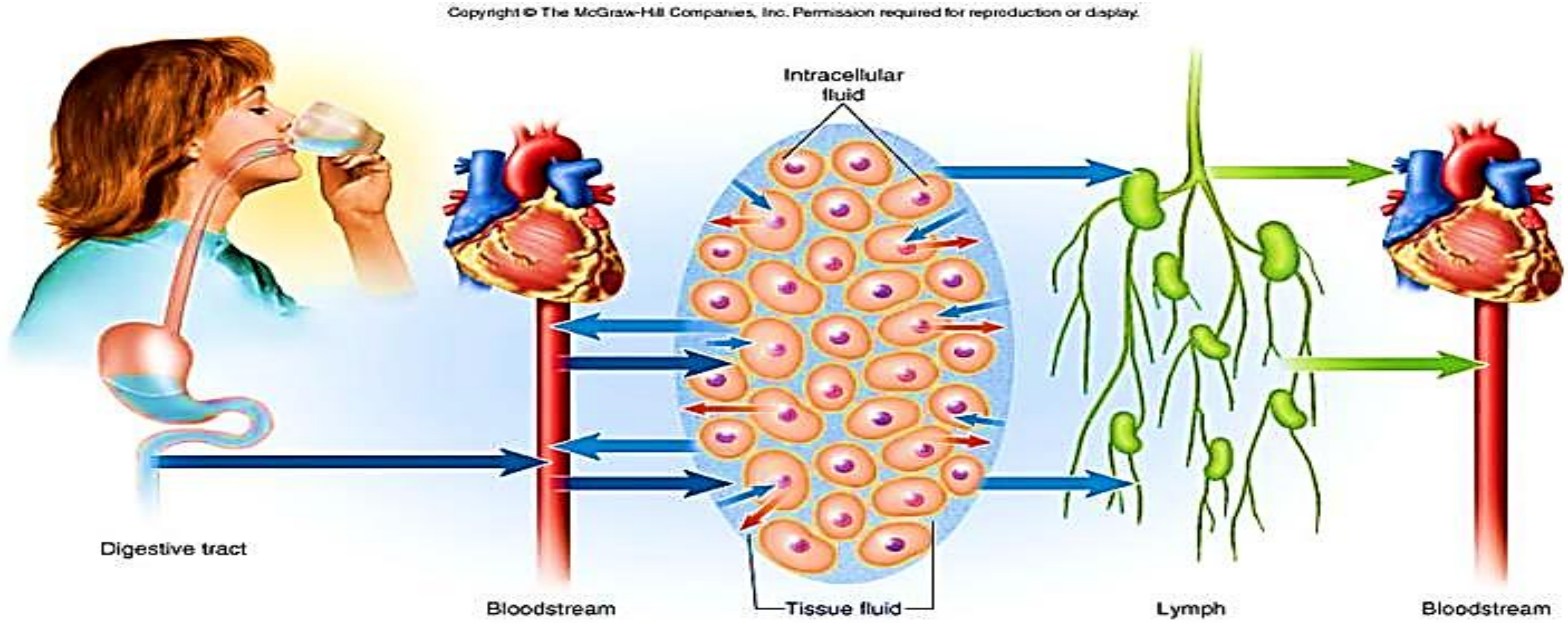
Fluid, Electrolyte and Acid-Base Balance



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Introduction

- In the human body, the substances that participate in chemical reactions must remain within narrows ranges of concentration. Homeostasis, or equilibrium of the internal environment, refers to the state of balance of body fluid; it is a fundamental property of all living things.

Functions of body fluids.

- Carry nutrients and waste.
- Facilitating digestion and promoting elimination.
- Act as a lubricant in joints.
- Give shape and form to the cells.
- Regulate body temperature.

Compartments of body fluids

The total body water is distributed primarily between 2 compartments, namely, extracellular fluid (ECF) and intracellular fluid (ICF) compartments.

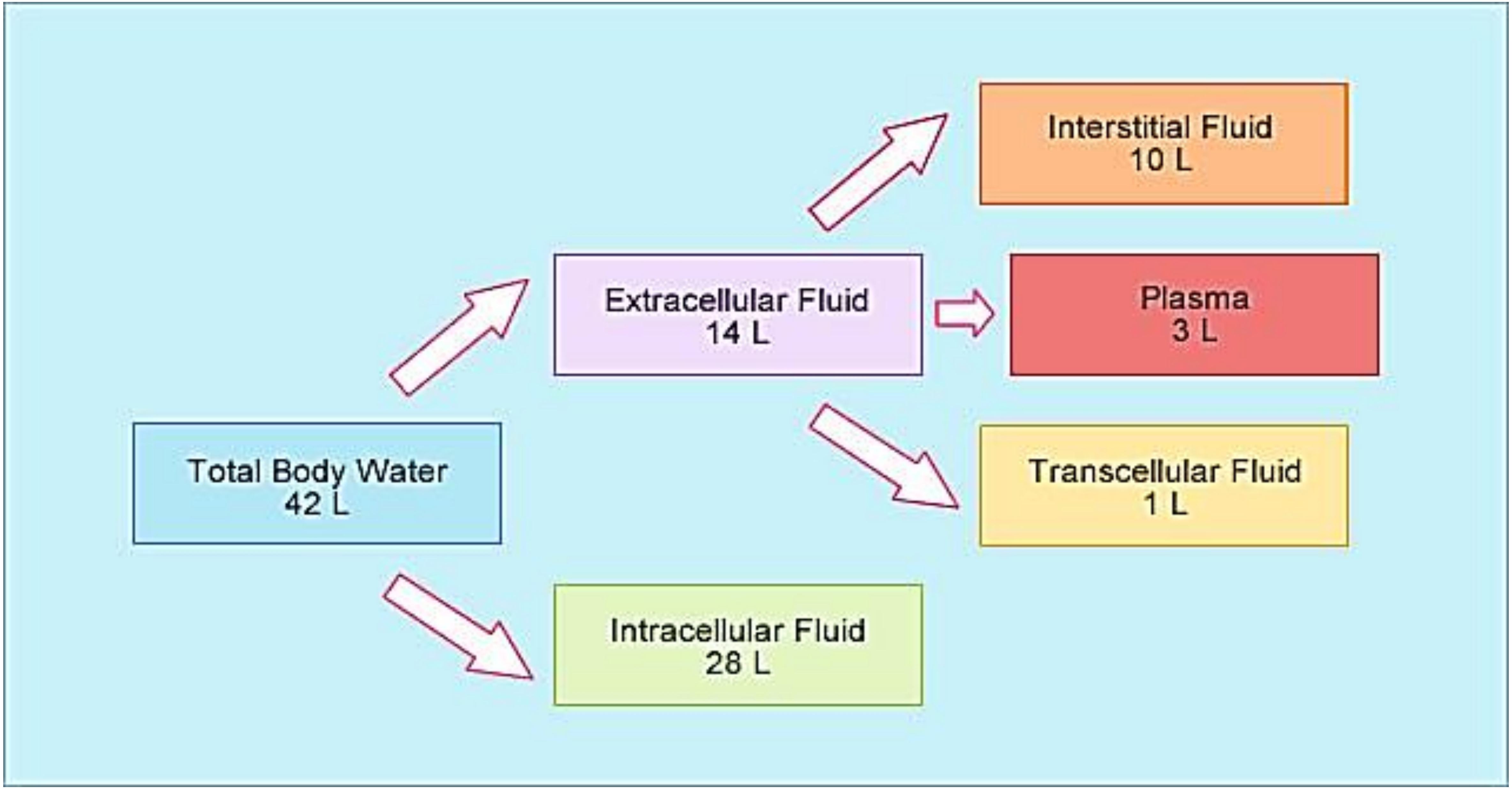
- **ECF is present outside cells:**
  - Makes up about ⅓ of total body water, 20% of total body weight and includes:
    - Intravascular fluid (¼ of ECF): primary component of plasma
    - Interstitial fluid (¾ of ECF): lies outside blood vessels.
    - Trans cellular (TCF)
- **ICF is present inside cells:**
  - Makes up about ⅔ of total body water and 40% of body weight.

Extracellular fluid (ECF):

- **The intravascular space** (IVF): plasma; ~1/4 ECF: 3 L of the average 6 L of blood volume is made up of plasma, the remaining 3 L is made up of erythrocytes, leukocytes, and thrombocytes.

- **Interstitial fluid (ISF):** between cells; ~3/4 ECF. The interstitial space contains the fluid that surrounds the cell. **Lymph is an interstitial fluid.**

- **Trans cellular (TCF):** is the smallest division of the ECF compartment and contains approximately 1 L of ECF. Examples of trans cellular fluids include cerebrospinal, pericardial, synovial, intraocular, and pleural fluids; sweat; peritoneal and digestive secretions.


Interstitial Fluid
10 L
Extracellular Fluid
14 L
Plasma
3 L
Total Body Water
42 L
Transcellular Fluid
1 L
Intracellular Fluid
28 L

Factors that influence on the amount body of fluid content:

1. **Age:** An infant especially on born prematurely has considerably more body fluid than an elderly person due to greater metabolic rate.

2. **Lean body mass:** Fat tissues contain a small amount of water whereas lean tissue is rich in water, the more obese the person, the smaller the percentage of total body water.

3. **Sex or gender**: Females tend to have proportionally more body fat than males; they also have less body fluid than males.

Factors that influence on the amount body of fluid content:

4. **Environment:** Excess heat stimulates the sympathetic nervous system and cause person to sweat.
5. **Diet:** In nutritional deficiency, the body preserved the protein, by breaking down the fat and glycogen.
6. **Stress:** Water retention and increase the production of antidiuretic hormone.
7. **Illness:** As burn by fluid and electrolytes loss, and as renal disorders.

Body water distribution and spacing:

1. Distribution varies with age, sex and body composition. Percentage of body weight that is **water is about 80% in a full-term infant**, **60% in a typical lean, adult male**, and **45-50% of obese and elderly**. This puts infants, elderly, and obese individuals at greater risk for fluid related problems.

2. Fluid spacing is a term used to classify the distribution of water in the body:

Fluid spacing

1. **First Spacing**: describes normal distribution of fluid in the body in both the intracellular and extracellular fluid compartments

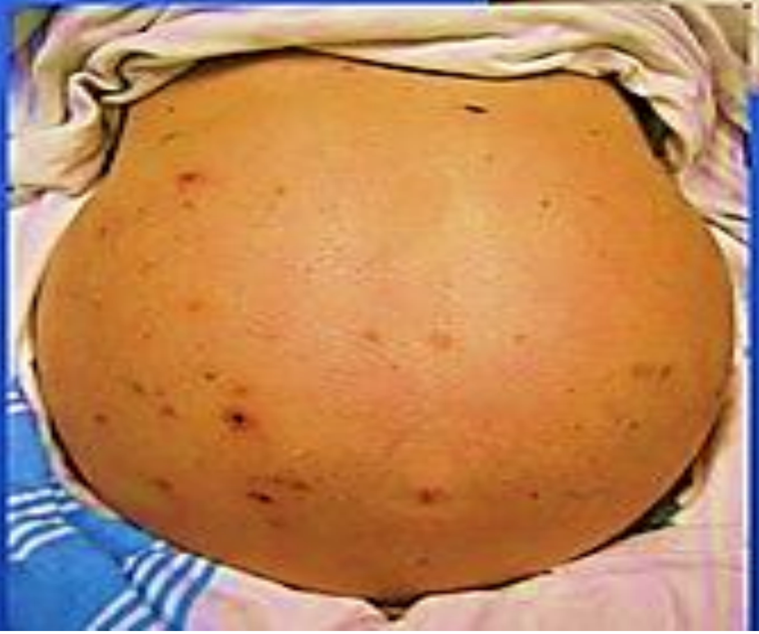
2. **Second Spacing:** describes the excess accumulation of fluid in the interstitial spaces, which we also call edema.

3. **Third Spacing:** occurs when fluid accumulates in areas that normally have no fluid or minimal amount of fluid, such as with ascites, and edema associated with burns. In extreme cases third spacing can cause a relative hypovolemia.

Fluid Status: 1 liter of water weighs 2.2 pounds. **A sudden weight gain or loss is the best indicator of fluid status.** Patients that need to have their fluid status **monitored should have not only their Intake and Output measured, but also daily bedside weights**.

Fluid Spacing

- First spacing
  - Normal
- Second spacing
  - Edema
- Third spacing
  - Ascites
  - Burn edema



Regulation of Body Fluid Compartments:

The physiological forces that affect the movement of body fluids through cell walls and capillaries can be perceived as a mass-transportation system that carries traffic between the compartments.

- Osmosis
- Diffusion
- Filtration
- Active transport

Osmosis

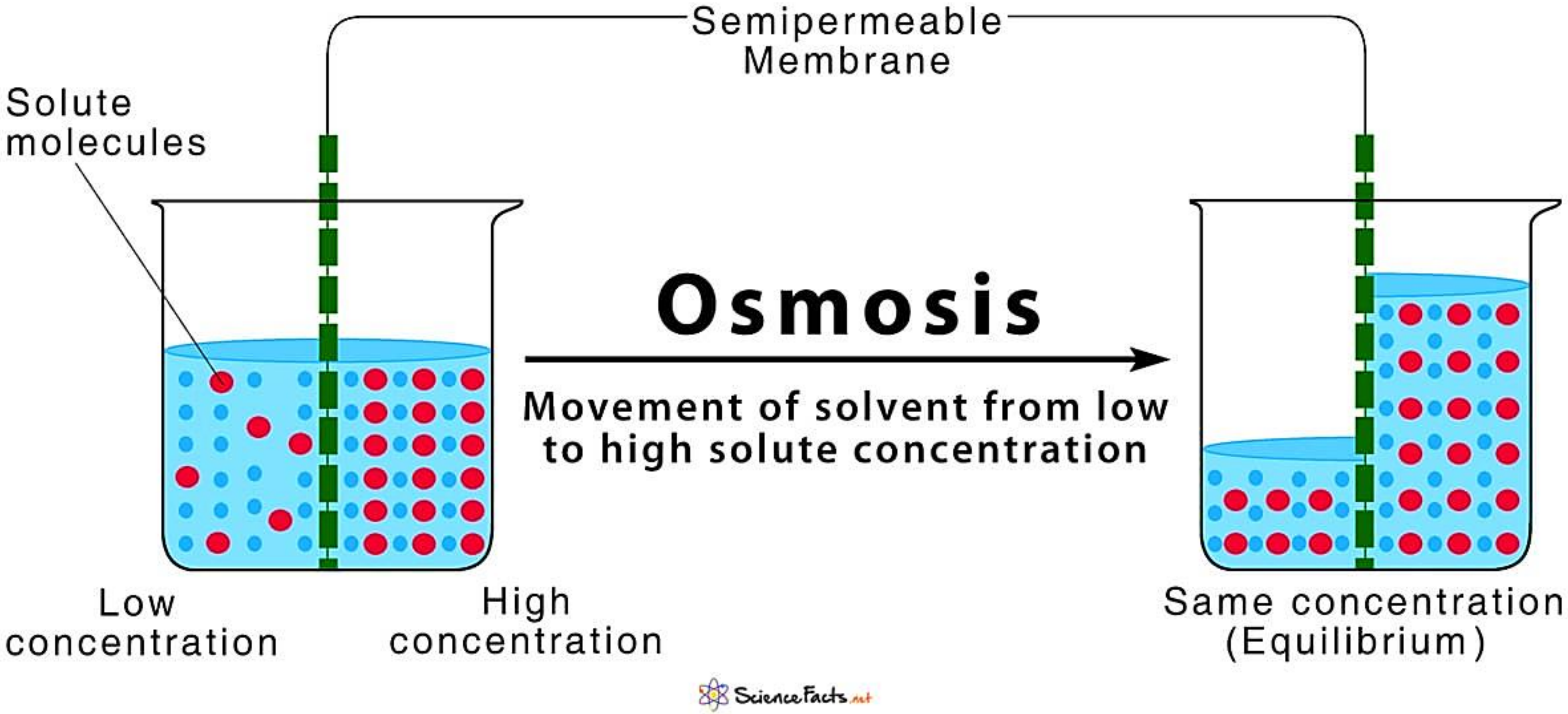
passage of a solvent from an area of lesser concentration to an area of greater concentration) is **influenced by:**

-The movement of water

-The semi permeability of the membrane .

Three other terms are associated with osmosis:

- **Osmotic pressure** is the amount of hydrostatic pressure needed to stop the flow of water by osmosis. It is primarily determined by the concentration of solutes.
- **Oncotic pressure** is the osmotic pressure exerted by proteins (eg, albumin).
- **Osmotic diuresis** is the increase in urine output caused by the excretion of substances such as glucose, mannitol, or contrast agents in the urine.

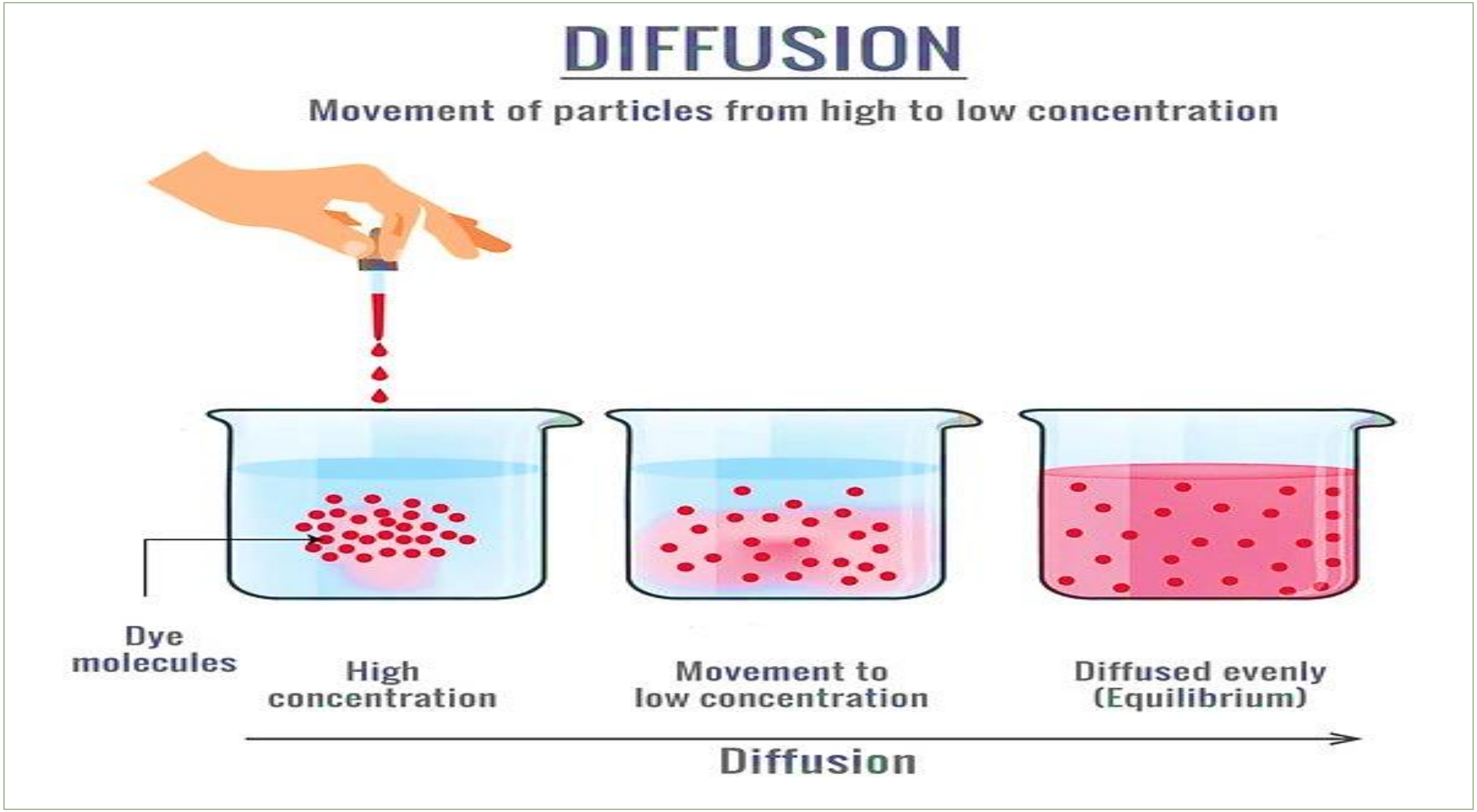

Semipermeable
Membrane
Solute
molecules
Osmosis
Movement of solvent from low
to high solute concentration
Low
concentration
High
concentration
Same concentration
(Equilibrium)
ScienceFacts.net

Diffusion

passive movement of **electrolytes** or other particles from areas of higher concentration to areas of lower concentration. **Examples of diffusion** are the exchange of oxygen and carbon dioxide between the pulmonary capillaries and alveoli.

The rate of diffusion is influenced by:

- The size of the molecule (smaller molecules diffuse faster than larger molecules).
- The concentration of the molecules (molecules move from an area of greater concentration to an area of lesser concentration).
- The temperature of the solution (higher temperatures increase the rate of diffusion).


DIFFUSION
Movement of particles from high to low concentration
Dye molecules
High concentration
Movement to low concentration
Diffused evenly (Equilibrium)
Diffusion

Filtration

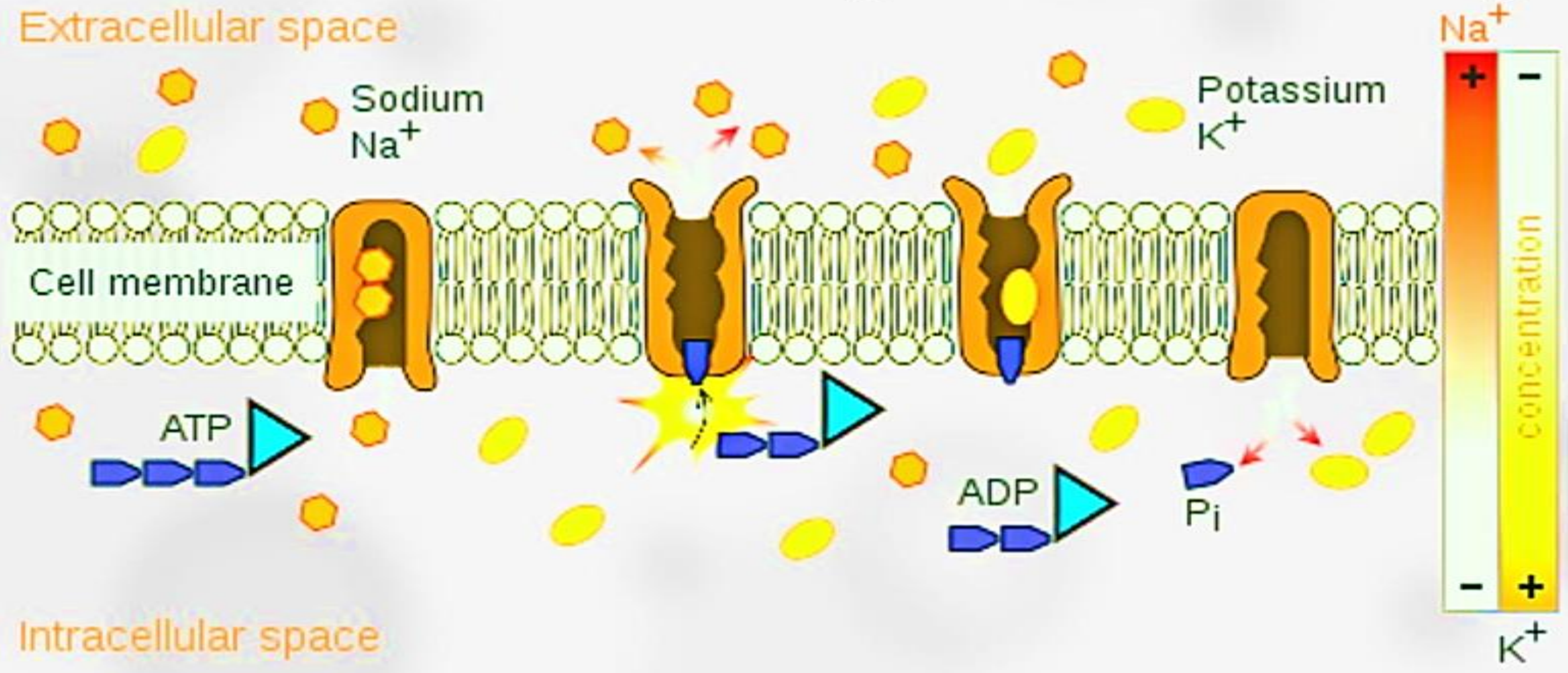
is the movement of water and solutes from an area of high hydrostatic pressure to an area of low hydrostatic pressure. Example for filtration is the kidneys filter approximately 180 L of plasma per day.

Active transport

Active transport requires energy in the form of adenosine triphosphate (ATP) to move electrolytes across cell membranes against the concentration gradient **(from areas of lower concentration to areas of higher concentration).** One of active transport is the **sodium-potassium pump, which moves Na+ out of a cell and K+ into it, keeping ICF lower in Na+ and higher in K+ than the ECF.**

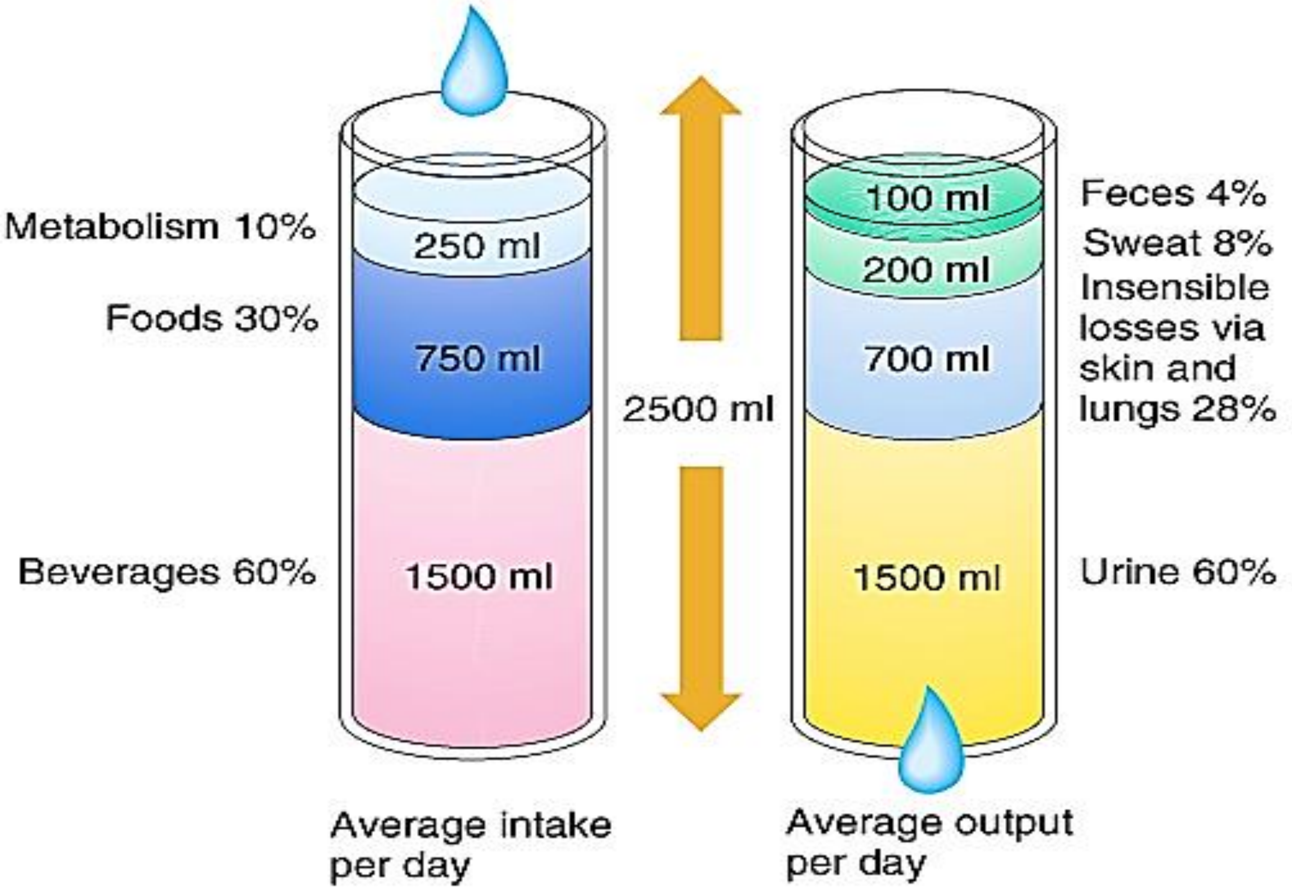
Active transport

Active transport is a type of cellular transport wherein substances move against the concentration gradient.



Fluid Intake and output in an adult (in 24 hours or / day)

In an otherwise healthy individual, daily fluid volume requirement estimations may be derived from calculations of weight (30-40ml/kg/day).



Regulators of Fluid Balance

- The body has many regulators that maintain fluid balance, including fluid and food intake, skin, lungs, gastrointestinal tract, and kidneys. When all organs are functioning normally, the body is able to maintain homeostasis.

Homeostatic Mechanisms

Human body include mechanisms that help regulate the body, this includes organs, glands, tissues and cells. The adjusting of these enables the body to constantly be in a steady state.

Organs and systems maintaining homeostasis

1. Kidneys: are the primary regulator of body fluids and electrolyte balance.

- They regulate the volume of ECF by regulating water and electrolyte excretion.
- The kidneys control the reabsorption of water from plasma filtrate and ultimately the amount excreted as urine. Although 135 to 180 L of plasma per day is normally filtered in an adult, only about 1.5 L of urine is excreted.
- Electrolyte balance is maintained by selective retention and excretion by the kidneys.
- The kidneys also play a significant role in acid–base regulation, excreting hydrogen ion (H+) and retaining bicarbonate.

Organs and systems maintaining homeostasis

2- Heart and Blood Vessel Functions;

the pumping action of the heart circulates blood through the kidneys under sufficient pressure to allow for urine formation. Failure of this pumping action interferes with renal perfusion and thus with water and electrolyte regulation.

3- Lung Functions;

They are also vital in maintaining homeostasis. Through exhalation, the lungs remove approximately 300 mL of water daily in the normal adult.

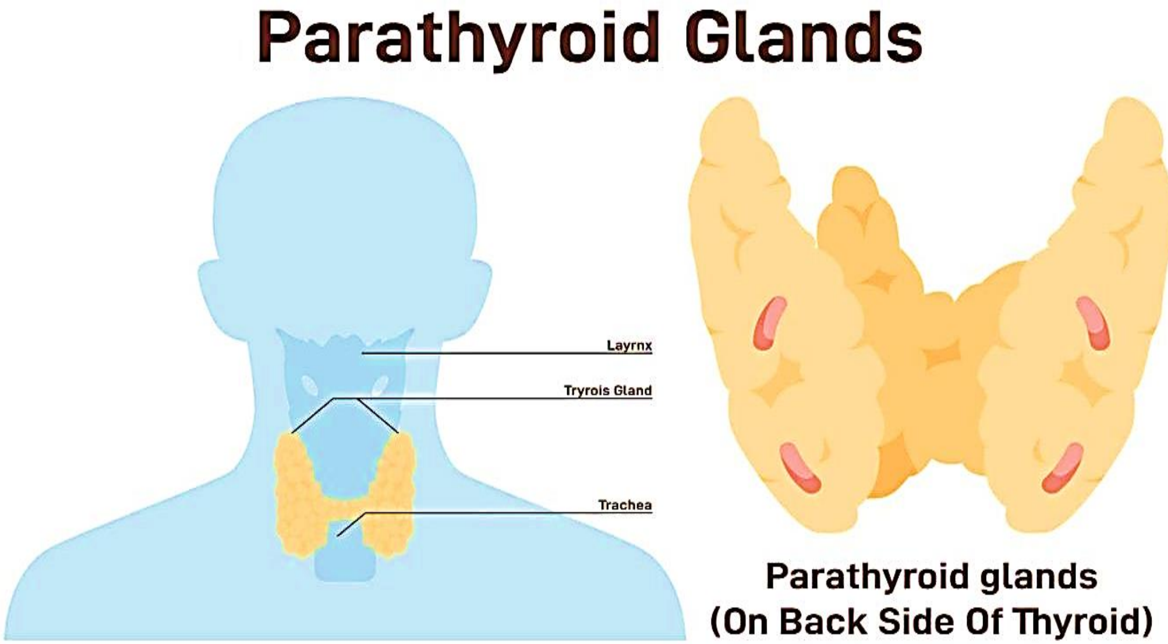
Organs and systems maintaining homeostasis

4- Pituitary Functions; The hypothalamus manufactures ADH, which is stored in the posterior pituitary gland and released as needed to conserve water. Functions of ADH include maintaining the osmotic pressure of the cells by controlling the retention or excretion of water by the kidneys and by regulating blood volume.

5- Adrenal Functions; Aldosterone, secreted by outer zone of the adrenal cortex, and effect on fluid balance. Increased secretion of aldosterone causes sodium retention, water retention) and potassium loss. Conversely, decreased secretion of aldosterone causes sodium and water loss and potassium retention., when Cortisol secreted in large quantities (or administered as corticosteroid therapy), it can also produce sodium and fluid retention.

Organs and systems maintaining homeostasis

6- Parathyroid Functions; they embedded in the thyroid gland, regulate calcium and phosphate balance by means of parathyroid hormone (PTH). PTH influences bone resorption, calcium absorption from the intestines, and calcium reabsorption from the renal tubules.



Electrolytes

- Are active chemicals (cations that carry positive charges and anions that carry negative charges).
- The major cations in body fluid are sodium, potassium, calcium, magnesium, and hydrogen ions.
- The major anions are chloride, bicarbonate, phosphate, sulfate.

- **Intracellular Fluid**: High concentrations of potassium, phosphate, and magnesium.
- **Extracellular Fluid:** High concentrations of sodium, chloride, calcium, and bicarbonate.

Common electrolytes

Electrolyte Ion	Normal range	Basic Functions	Dietary Sources
Sodium (Na^+)	135-145 mEq/L) (mmol/L).	• Transmitting nerve impulses and contracting muscles	Table salt, cheese, meat, poultry, shellfish, fish, eggs.
Potassium (K^+)	3.5 to 5 mEq/L	• Regulating cardiac impulse transmission and muscle Contraction. • Skeletal and smooth muscle function. • Regulating acid-base balance.	Fruits, especially bananas, oranges, and dried fruits; vegetables, meats, and nuts.
Calcium (Ca^{2+})	8.5-10 mg/dl	• Forming bones and teeth • Transmitting nerve impulses. • Regulating muscle contractions. • Maintaining cardiac pacemaker (automaticity) • Blood clotting.	Dairy products (milk, cheese, and yogurt), sardines, whole grains, and green leafy vegetables.
Magnesium (Mg^{2+})	1.3 to 2.1 mEq/L	• Relaxing muscle contractions. • Transmitting nerve impulses. • Regulating cardiac function.	Green leafy vegetables, whole grains, fish, and nuts.
Chloride	95 to 108 mEq/L	• HCl production. • Regulating acid-base balance. • Buffer in oxygen-carbon dioxide exchange in RBCs.	In table salt or sea salt as sodium chloride. Foods with higher amounts of chloride include seaweed, tomatoes, lettuce, olives
Phosphate	2,5 to 4.5 mEq/L(mmol/L). .	• Forming bones and teeth. • Metabolizing carbohydrate, protein, and fat. • Regulating calcium levels	Milk and milk products and meat and alternatives, such as beans, lentils and nuts. Grains, especially whole grains provide phosphorus. Phosphorus is found in smaller amounts in vegetables.
Bicarbonate	22 and 26 mEq/L(mmol/L).	Major body buffer involved in acid-base regulation.	Bicarbonate (HCO_3) may be present in foods or medicines as part of sodium bicarbonate, potassium bicarbonate

Terms for electrolytes disturbances

Electrolyte	Name if increased	Name if decreased
Sodium (Na^+)	Hypernatremia	Hyponatremia
Potassium (K^+)	Hyperkalemia	Hypokalemia
Calcium (Ca^{2+})	Hypercalcemia	Hypocalcaemia
Magnesium (Mg^{2+})	Hypermagnesaemia	Hypomagnesaemia
Phosphate	Hyperphosphatemia	Hypophosphatemia
Chloride	Hyperchloremia	Hypochloremia
Bicarbonate	Metabolic Alkalosis	Metabolic Acidosis

Nursing assessment for fluid balance

- **Three elements for Assessing fluid balance:**

I- Clinical assessment

II- Review of fluid balance charts

III- Review of blood chemistry

I- Clinical assessment

- Patients should be asked if they are thirsty.
- Assess vital signs.
- Assess capillary refill time (CRT) is a good measure of the fluid present in the intravascular fluid volume, it is measured by **holding the patient's hand at heart level and pressing on the pad of their middle finger for five seconds**, the pressure is released, and the time measured in seconds until normal color returns. Normal filling time is usually less than two seconds.
- Assessing skin turgor is a quick and simple test performed by pinching a fold of skin.
- Measuring Body weight, is a good indicator of hydration status.

II-Review of fluid balance charts

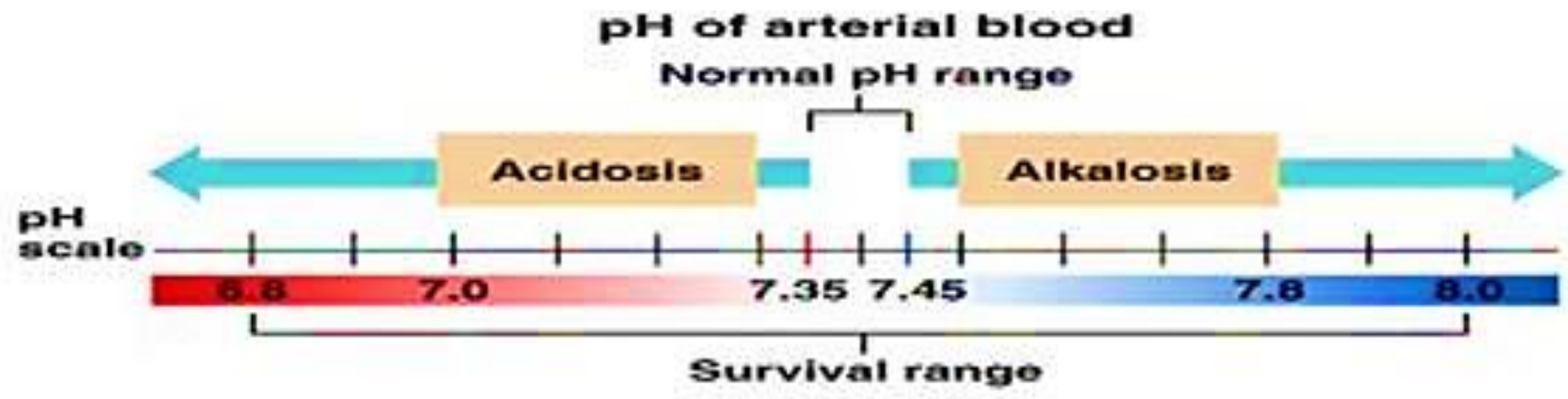
- **Normal urine output is around 1ml/kg of body weight per hour**, in a range of 0.5-2ml/kg per hour. Monitoring a patient's fluid balance to prevent dehydration or overhydrating.

III.Review of blood chemistry. The analysis of blood chemistry as sodium, potassium, chloride, bicarbonate, are essential to measure hydration status.

- **Blood urea nitrogen:** the end result of protein metabolism. Normal BUN is 10 – 20mg/dl.
- **Urine specific gravity:** measure the ability of the kidney to secrete or conserve urine. Normal specific gravity. 10.10 - 10.25.
- **Creatinine:** indicate the function of the kidney. Normal Creatinine in female is 0.5- 1.1 mg/dl while for male is 0.6- 1.2
- **Urine output:** In healthy people, urine should be a pale straw color. It should be clear, with no debris or odor.

Acid-Base Balance

- Refers to the homeostasis of the hydrogen ion concentration in extracellular fluid. The pH symbol is used to indicate the hydrogen ion concentration of body fluids; 7.35 to 7.45 is the normal pH range of extracellular fluid.
- Hydrogen ions (H+), which carry a positive charge, are protons. Depending on the number of hydrogen ions present, a solution can be acidic, neutral, or alkaline.
- **PH**: is the concentration of hydrogen ion (H+) in the blood. Its normal range is 7.35 to 7.45.



Acid-Base Balance

- **An acid** is a substance containing hydrogen ions that can be liberated or released.
- **An alkali** is a base, which is a substance that can accept or trap a hydrogen ion.
- A solution becomes more acid; the pH becomes less than 7.
- Gastric secretions that are strongly acidic have an approximate pH of 1 to 1.3, whereas strongly alkaline pancreatic secretions have an approximate pH of 10.
- **Acidosis** pH falls below 7.35.
- **Alkalosis** pH exceeds 7.45.

Normal ranges of Arterial blood Gase(ABG)

Variable	Normal Range
PH	7.35 - 7.45
paCO2	35-45 mmhg
Pao2	Above 90mmhg
Bicarbonate (HCO3 –	22-26 mEq/L (22-26 mmol/L)

Acid-Base disturbance

- **Paco2:** is the partial pressure of co2 dissolved in arterial plasma. Its normal range is 35-45mmhg.
- Increase Paco2 than 45 lead to **respiratory acidosis** while decrease it than 35 lead to **respiratory alkalosis**.
- **Hco3:** is the concentration of Sodium bicarbonate in the blood. Its normal range is 22-26 meq/L
- Increase Hco3 than 26 lead to **metabolic alkalosis** while decrease it than 22 lead to **metabolic acidosis**.
- **Pao2:** is the partial pressure of o2 dissolved in arterial plasma. Its normal range is 75-100 mmhg

Types of Acidosis and Alkalosis

	pH	pCO_2	Total HCO_3^-
Metabolic acidosis	↓	N, then ↓	↓
Respiratory acidosis	↓	↑	N, then ↑
Metabolic alkalosis	↑	N, then ↑	↑
Respiratory alkalosis	↑	↓	N, then ↓

Arterial Blood Gases

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PH acidosis < 7.4 > alkalosis

PH	7.35 – 7.45
$PaCO_2$	35 – 45
HCO_3	22 – 28

← Respiratory

← Metabolic

R Respiratory — PH ↑ PCO_2 ↓ Alkalosis

O Opposite — PH ↓ PCO_2 ↑ Acidosis

M Metabolic — PH ↑ HCO_3 ↑ Alkalosis

E Equal — PH ↓ HCO_3 ↓ Acidosis

Uncompensated : CO_2 or HCO_3 normal

Partially Compensated : Nothing is normal

Compensated : PH is normal (7.4 baseline/neutral)

Thank you

"Thank you for being the reason I smile every day."